



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2018

Bachelor in Computer Application (BCA)
Course Title: Computer Graphics and Animation
Code No: CACS 305
Semester: V
Symbol No:

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Center:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt All Questions:

[10×1=10]

1. Circle (O) the correct answer in the following questions:

- a) Random scan system is used for
 - i. Color drawing application
 - ii. Line drawing application
 - iii. Pixel drawing application
 - iv. All of the above
- b) Which of the following is used to store picture definition in random display system?
 - i. Display Controller
 - ii. Vertical retrace
 - iii. Refresh display file
 - iv. Frame buffer
- c) Which of the following algorithm is used to fill the interior of polygon?
 - i. Parallel curve algorithm
 - ii. Flood fill algorithm
 - iii. Scan line fill algorithm
 - iv. Boundary fill algorithm
- d) Which of the following equation is used in 2D scaling of point $P(x,y)$ to $P'(x',y')$
 - i. $x'=x*sx$ and $y'=y*sy$
 - ii. $x'=y*sx$ and $y'=x*sy$
 - iii. $x'=x+sx$ and $y'=y+sy$
 - iv. $x'=x+sy$ and $y'=y+sx$
- e) What is the value of initial decision parameter in the midpoint circle drawing method?
 - i. $\frac{4}{5}-r$
 - ii. $\frac{3}{5}-r$
 - iii. $\frac{5}{3}-r$
 - iv. $\frac{5}{4}-r$
- f) Which of the following hidden surface removal algorithm does not employ image space approach?
 - i. Depth Buffer method
 - ii. A buffer method
 - iii. Back face detection method
 - iv. Scan line method
- g) In Cohen Sutherland line clipping, 4 bit region code of bottom left region of window is:
 - i. 1100
 - ii. 0101
 - iii. 101
 - iv. 1001
- h) Which of the following is the space in which application model is defined?
 - i. World coordinate system
 - ii. Screen coordinate system
 - iii. Vector coordinates system
 - iv. Interface window
- i) The intersection of primary colors in CMYK color model generates:
 - i. Green color
 - ii. Blue color
 - iii. Dark red color
 - iv. Black color
- j) Which of the following allows you to insert text within your flash stage?
 - i. Text box
 - ii. Text tool
 - iii. Pen tool
 - iv. Oval tool



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Group B

Attempt any six questions:

[6×5=30]

2. What do you mean by area filling? Explain boundary fill algorithm.
3. Define scaling transformation? Prove that two successive scaling are multiplicative.
4. What is symmetric property of circle? Digitize a circle whose center is at origin and radius is 9.
5. Define window and view port. Explain 2D viewing transformation pipeline.
6. Reflect a prism A(0,0,0), B(1,1,0), C(1,2,2) and (0,2,0) about yz-plane which has been rotated previously with +90 degree about y-axis.
7. What do you mean by visible surface detection? Explain Z buffer algorithm for visible surface detection.
8. What is animation? Explain the steps to create animation sequence.

Group C

Attempt any two questions:

[2×10=20]

9. Define refresh rate and persistence of monitor. Explain architecture of random scan display system with block diagram.
10. How can you draw line using Bresenham's line drawing algorithm? Trace it to rasterize the line with end points P (1, 4) and Q (7, 8).
11. Explain Cohen Sutherland line clipping algorithm. Let ABCD be the rectangular window with A(20,20), B(90,20), C(90,70) and D(20,70). Use Cohen Sutherland algorithm to find the region codes and clip the lines with end point P(10,30) and Q(80,80)



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2021

Bachelor in Computer Applications
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Code No: CACS305
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2019

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

- i) Random scan systems generally use ____ to display an image
a) Memory b) Frame buffer c) Processor d) Refresh Display File
- ii) Which of the following is used for the area of the computer captured by an application?
a) Viewport b) Display c) Window d) Buffer
- iii) Which color is produced by the intersection of RGB
a) Blue b) White c) Green d) Red
- iv) In the back-face detection algorithm, if V is along positive Z direction than polygon is Back Face if.....
a) $C \geq 0$ b) $C \leq 0$ c) $C \geq 1$ d) $C \leq 1$
- v) How many steps are required to scale an object about a fixed point?
a) 1 b) 2 c) 4 d) 3
- vi) Which types of lines are ignored in the scan line algorithm to fill regions?
a) Horizontal b) Vertical c) Left Inclined d) Right Inclined
- vii) Reflection of a point about X-axis followed by a counter clockwise rotation of 90 degree is equivalent to reflection about the line?
a) $x = -y$ b) $y = -x$ c) $x = y$ d) $x + y = 1$
- viii) What is the 4-bit code of the top left-region among the nine regions divided using the Cohen-Sutherland algorithm?
a) 0001 b) 1001 c) 0110 d) 0000
- ix) Which of the following equation is used in 2D scaling to scale a point(x,y) to the new point (x',y')?
a) $x' = x \times s_x$ and $y' = y \times s_y$ b) $x' = x + s_x$ and $y' = y + s_y$
c) $x' = x \times s_y$ and $y' = y \times s_x$ d) $x' = x + t_x$ and $y' = y + t_y$
- x) What part of flash menu bar allows you to hide/unhide panels?
a) View b) Edit c) Window d) Layer Panel



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2021

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Pass Marks: 24
Time: 3 hours
Batch: 2019

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Define horizontal and vertical retrace of electron beam. Explain architecture of random scan display system. [2+3]
3. How does decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center has (10, 5) using mid-point circle algorithm. [1+4]
4. What do you mean by window to viewport transformation? Explain 2D viewing pipeline. [2+3]
5. Generate the points between the end points of a line viz. (2, 2) and (9,6) by using DDA line drawing algorithm.
6. Define visible surface detection. Outline the Z-buffer algorithm along with its advantages and disadvantages. [1+4]
7. What do you mean by Virtual Reality? Explain the types of Virtual Reality. [2+3]
8. Define reflection transformation and derive the 2D reflection matrix along x-axis and y-axis in homogeneous coordinate. [1+4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What is the need of homogeneous coordinate system in geometric transformation system? Find the new co-ordinate of rectangle ABCD whose center is at (4, 2) is reduced to half of its size and center will remain same. The co-ordinate of ABCD are A(0, 0), B(0, 4), C(8, 4) and D(8, 0). [2+8]
10. Explain Sutherland Hodgeman Polygon Clipping algorithm and trace it to clip a polygon with end points P(10, 30), Q(25, 35), R(35,50), S(60,30), T(45, 28), U(40,15) against a window whose lower left corner is at (15, 25) and upper right corner is at (50, 40). [4+6]
11. What do you understand by object space and image space method? Explain Flood Filled algorithm with its applicability and concept. [4+6]



Tribhuvan University
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2024

Bachelor in Computer Applications

Course Title: Computer Graphics and Animation

Code No: CACS 305

Semester: V

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Differentiate between beam penetration method and shadow mask method. How does raster scan display system work? [3+2]
3. How DDA works? Digitize the line segment with the endpoints (15, 25) and (21, 35) using DDA line drawing algorithm. [2+3]
4. What is a viewport? Consider a window with lower left corner at (2, 2) and upper right corner (5, 10) and a viewport with left lower corner at (5, 5) and upper right corner at (8, 8). What will be the value of the point in the viewport after the window to viewport transformation if the point is (4, 4) in the window? [1+4]
5. Explain how scan line algorithm can be used for hidden surface removal. [5]
6. Derive the transformation matrices for 3D rotation and reflections. [2.5+2.5]
7. What is animation? Explain about applications of Virtual Reality. [1+4]
8. Write short notes on (any two): [2.5+2.5]
 - a) RGB color model
 - b) 2D shear transformation
 - c) Computer Graphics Vs. Image Processing

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Differentiate between boundary fill algorithm and flood fill algorithm in detail. Find the composite transformation matrix for anti-clockwise rotation of 60° about a point (2, 3). Use it to rotate a triangle ABC with vertices A (4, 3), (5, 5) and (8, 9). [5+3+2]
10. What is visible surface detection? How object space method can be used for visible surface detection? Explain depth-sorting method for visible surface detection in detail. [2+2+6]
11. How region codes are used in Cohen-Sutherland algorithm to clip the line segments? Find the clipping coordinates for lines PQ and RS with P(20, 5), Q(40, 30) and R(20, 20), S(80, 60) against the window with left upper corner(10, 50) and right lower corner(50, 10). [4+6]

Tribhuvan University
Institute of Science and Technology
2020

Bachelor Level / Fifth Semester / Science

Computer Science and Information Technology(CACS305)

((TU BCA) Computer Graphics and Animation)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Full marks: 60

Pass marks: 24

Time: 3 hours

Group B

Attempts any SIX questions.

[6x5=30]

2. What is computer graphics? Explain different application areas of computer graphics.
3. How can you draw a circle using mid-point circle algorithm? Explain with the algorithm.
4. Explain 3D basic geometric transformation with an example.
5. What is polygon clipping? Explain Sutherland Hodgman algorithm for polygon clipping.
6. Given a triangle with vertices A(2,3), B(5,5), C(4,3) by rotating 90 degree about the origin and then translating two unit in each direction. Use homogenous transformation matrix to find the new vertices of the triangle.
7. Explain the scan line algorithm for visible surface detection.
8. Explain the architecture of VR system with necessary components.

Group C

Attempts any TWO questions.

[2x10=20]

9. Explain the working details of DDA line drawing algorithm? Digitize the line with endpoints (2,2) and (10,5) using Bresenham's line drawing algorithm. [5+5=10]
10. Write the different between object space method and image space method. Explain Z buffer algorithm for visible surface detection. [5+5=10]
11. Derive the formula for windows to viewport transformation. Given a window bordered by (0,0) at the lower left and (4,4) at the upper right. Similarly, a viewport bordered by (0,0) at the lower left and (2,2) at the upper right. If a window at position (2,4) is mapped into the viewport. What will be the position of viewport to maintain same relative placement as in window? [5+5=10]